

What Do People Optimize in Graph Layout?

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Abstract

Automatic layout tools place the boxes of a diagram by minimising a weighted sum of aesthetic criteria: few edge crossings, no overlapping boxes, edges of similar length. If people drew the same way, a diagram laid out by hand would be a local minimum of that sum and a tool started from it would find nothing to change. We test each criterion separately on 853 diagrams of 15 to 40 boxes whose coordinates a person chose: 305 WikiPathways biological pathways, 248 Reactome SBGN maps and 300 BPMN process models, three communities with different authoring tools. The test is direct: move each box 2% of the drawing width in each of sixteen directions and ask whether any move lowers the criterion, edges taken as the polylines the files store; the same test runs on layouts of the same graphs by four tools, **neato**, **neato** with overlap removal, **dot** and ELK layered. Under overlap every box of the median hand diagram is held at term value zero, as in every tool layout that removes overlaps. Under uniform edge length and under stress no hand-placed box is held in any corpus, while **neato**'s layouts are held under stress at 1.00, so the test has power; the three-term energy tools use, which at the hand layout is 99% edge length by value, holds none of it, and no reweighting repairs it: a weight of 0.001 on length or stress drops the held fraction below 0.015 in every corpus. Alignment of boxes into shared rows and columns holds 0.52, 0.21 and 0.91 of hand-placed boxes, against 0.06 in **neato**'s layouts, 0.89 to 1.00 in **dot**'s and 0.42 to 0.62 in ELK's. The BPMN lanes do not explain it: 43% of the held boxes are held by a column or by a row shared across lanes, which a horizontal lane cannot impose, and the lanes of the rest are a median 6.3 box heights tall, room the person did not use. Orthogonality on the drawn routes holds 0.73 of BPMN boxes; reading direction holds 0.59 to 0.85 by hand against 1.00 for the layered tools; crossings hold every box in every layout at this radius, the diagrams being near-trees (median edges per box 1.06, 1.04 and 1.11 against 1.30 for the Rome benchmark) with no crossing to remove on the median hand diagram's routes. On every criterion the hand profile is the layered tools': held under overlap, free under length and stress, aligned in part where **dot** is exact. The measurement is deterministic, and an uncapped descent from each hand layout confirms the test's power, ending held at 1.00 under every energy it reports on.

1 Introduction

A diagram is a graph with a drawing: boxes for the nodes, lines for the edges, and a position for every box. When a program chooses the positions it minimises an *energy*, a weighted sum of quantities the field has agreed are bad: edges that cross, boxes that overlap, edges of very unequal length, boxes that sit on edges they do not belong to [3, 2, 4, 5]. The weights are asserted by the tool's author. The premise is that a person drawing by hand is doing the same thing more slowly, so that the drawing they accept is one a tool would also accept.

The premise has a direct test, one criterion at a time. Take a diagram a person drew. Move one box a small distance in each of several directions and ask whether any of the moves lowers

*Independent researcher. Code, parsed corpora and every table's script: <https://github.com/Anode1/cjitter>, directory `example/diagrams`.

the criterion. If none does, the box is *held*: the person has put it where the criterion wants it, or where the criterion has stopped caring. Do this for every box and the fraction held says how much of the layout is a local minimum of that criterion; the criterion’s value says how well it is met. A criterion that holds every box at value zero is one the person satisfied; one that holds no box is one the person did not consider, or traded away.

We ran the test on 853 diagrams of 15 to 40 boxes from three unrelated communities (Section 4), and on layouts of the same graphs by four tools. The findings:

1. Under overlap every box of the median hand diagram is held and the term is zero in every corpus (the term is nonzero in 4 to 8% of hand diagrams). Every box in a layout by a tool that removes overlaps is held too; *neato* without overlap removal holds 0.41 to 0.76. Overlap is a criterion people satisfy exactly, in common with every tool that tries.
2. Under uniform edge length and under stress no hand-placed box is held: 0.00 in every corpus under either. The test has power: *neato*’s stress layouts are held under stress at 1.00 in every corpus. Under the three base criteria summed with the usual equal weights the length term carries 99% of the value and no box is held, which is what an earlier version of this study, using a capped descent, reported as “hand layouts are not stationary”. The weight sweep says why no reweighting repairs it: a weight of 0.001 on length or stress, the rest on crossings and overlap, drops the held fraction of nodes from 0.94–0.98 to below 0.015 in every corpus. Any positive weight on a criterion that holds no box makes the sum hold no box.
3. Alignment of boxes into shared rows and columns, HOLA’s gridiness, puts 58%, 36% and 82% of hand-placed boxes in an alignment of three or more; *neato* puts none, *dot* 88%, 82% and 78%, ELK 80%, 69% and 75%. The continuous form A1 holds 0.52, 0.21 and 0.91 of hand-placed boxes against 0.06 for *neato*, 1.00, 1.00 and 0.89 for *dot* and 0.57, 0.42 and 0.62 for ELK. The BPMN lanes do not explain the 0.91: of the held boxes, 30% are held by a column and 13% by a row shared with a box of another lane or of none, neither of which a horizontal lane can impose, and the rows within one lane sit in lanes a median 6.3 box heights tall. Orthogonality of the drawn routes holds 0.19, 0.17 and 0.73, more than any tool; separation of boxes from foreign edges is nearly satisfied everywhere (value 0.000 to 0.004) and holds 0.88 to 1.00, the flatness of a term with little to act on. In WikiPathways 0.37 of boxes share an exact x with another box while 17% of coordinates are integers, so the alignment there is the person’s act.
4. Reading direction does not close the gap: the flow term holds 0.59, 0.62 and 0.85 of hand-placed boxes at values 0.13, 0.10 and 0.05, where *dot* and ELK hold 1.00 at 0.000 in every corpus. The person tolerates a backward edge the layered tools remove.
5. On every criterion but alignment and flow the hand layout’s profile is the layered tools’: held under overlap, free under length and stress, partly held under orthogonality. The person draws like a layered tool that aligns part of the drawing and points most of it one way.
6. Crossings hold every box in every layout at radius 0.02, because no single move of that length removes one, so the count is flat. On the drawn routes the median hand diagram has no crossing in any corpus (on centre-to-centre chords WikiPathways reads one per 26 edges); *dot* leaves a median 0.029 and 0.056 per edge on the two biological corpora, ELK 0.036 and 0.066, both 0.000 on BPMN. The diagrams people draw are near-trees, median edges per box 1.06, 1.04 and 1.11 against 1.30 in the Rome benchmark [16]; crossings are what the theory is about [1], and on these diagrams there are few to remove.

The rest of the paper is the foundation behind the test (Section 2), the instrument (Section 3), the corpora, the method, the measurements, the alternatives not yet excluded (Section 7), and what is already known (Section 8). A tool builder reading only this far has the changes: take the weight off edge length, add alignment, and print the fraction of boxes the energy holds when started from a layout a person accepted.

2 Foundations

The energy. A layout of a graph with N nodes is a point $x \in \mathbb{R}^{2N}$, two coordinates per node, the boxes' sizes held fixed. An aesthetic energy is

$$E_w(x) = \sum_{k=1}^K w_k T_k(x), \quad w_k \geq 0, \quad (1)$$

where each term T_k is a quantity to be made small. The tool's author asserts the w_k ; the field of multicriteria layout is the choice of the T_k and the w_k [2, 4].

Local minimum. A point x is a local minimum of E if there is a radius $r > 0$ with $E(y) \geq E(x)$ for every y within distance r of x . For a differentiable E this implies $\nabla E(x) = 0$. Two of our terms are not differentiable: crossings is a count, piecewise constant in x , and overlap has corners where boxes touch. So the test uses no gradient. Fix a radius d and a set V of unit directions in the plane (sixteen, equally spaced). For node i and direction v let $x + dv^{(i)}$ be the layout with node i moved by d along v and every other node fixed. Node i is *held* at radius d if

$$E(x + dv^{(i)}) \geq E(x) \quad \text{for every } v \in V, \quad (2)$$

and the estimand is

$$q_d(x) = \frac{1}{N} \#\{i : \text{node } i \text{ is held at radius } d\}. \quad (3)$$

$q_d = 1$ says the layout is a local minimum of E over single-node moves of length d ; for smooth E and small d it says $\nabla E(x) = 0$ to within the resolution of V . Because (2) is a comparison of two energies it is defined for every term, and because it is evaluated per term it separates the criteria a person satisfied from those they did not. A descent-based measure (how far a capped descent moves the layout, the estimand of the first version of this study) answers the same question for the sum and cannot separate the terms; it is retained as a secondary.

The inverse problem. Given layouts $x^{(1)}, \dots, x^{(G)}$ that people accepted, inverse optimisation asks for the weights w under which they are nearest to stationary [18, 19]. With the directional differences $D_{iv}T_k = T_k(x + dv^{(i)}) - T_k(x)$ in hand, node i is held under w exactly when $\sum_k w_k D_{iv}T_k \geq 0$ for all v , a set of linear inequalities in w . The residual is the smallest total violation over w on the simplex, a linear program, and it is zero only if some weighting of the terms makes the layout a local minimum. A term with $D_{iv}T_k < 0$ for some v at almost every node (uniform edge length, at any layout whose edges are not all the same length) cannot take positive weight without violating the inequalities at those nodes, which is the zero weight the random-search fit found.

Step terms. A term that is a count (crossings) or a step function of position (gridiness, Section 5) is flat almost everywhere, so it holds a node wherever no single move of length d changes it. For such a term q_d says how much of the layout the criterion has stopped acting on, which is a different thing from how well the criterion is met, and the term's value at the layout is reported beside q_d .

The reference length. The length and stress terms compare distances with a reference L . Fixed at the layout’s median edge length, L is in general not the scale at which the term is least for that layout, so at a true minimum every node still has a scaling direction that lowers the term and the converged reference is not held (neato’s stress layouts: $q_{0.02} = 0.07$ to 0.12). L is therefore the value at which the term is least for the layout as loaded, $\sum \ell^2 / \sum \ell$ over edges for length and the same over $\|x_i - x_j\| / \delta_{ij}$ for stress, fixed before any move; neato’s stress layouts are then held at 1.00 in every corpus. The median and $1/\sqrt{N}$ are sensitivities.

The null and the controls. A test that can only succeed is not a test. The *stationarity reference*: q_d on a layout a descent has converged on must be near one, or the test lacks power for that energy; neato’s layouts are the reference for stress (1.00 in every corpus) and the uncapped climber’s end point for the sums (median 1.00 for every energy and corpus, Section 6). The *tool controls*: q_d on layouts of the same graph with the same box sizes by **neato** (stress, boxes left where they overlap), **prism** (neato followed by its overlap removal) and **dot** (layered), which say what a layout that does minimise something scores on each term. The *matched-budget control*: uniform random placement at the same number of energy evaluations, which the hand layout must beat, or “held under overlap” would be a property of sparse graphs and not of the person.

3 The instrument

The descents, the control and the paired statistics come from **cjitter**, a C library of four stochastic searches over a box of real variables: hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform sampling as the control. Every run is byte-reproducible from its seed on every platform, every method constant is a visible field, and **cjitter_compare** runs all four at the same budget on the same seeds and reports an exact paired sign test, Holm-corrected, against the control. “Better than uniform sampling at equal cost” is the only claim it lets a search make.

It is the latest of a series of small tools by the author, each reporting the condition under which its own answer is wrong:

year	tool	what it reports against itself
1997	backpropagation thesis [32]	descent on an error energy, derived
2001	label placer	per-label random steps, kept if the summed overlap fell
2011	least squares, now linearr	when a line is the wrong shape
2025	bpnn	when the network has memorised the rows
2026	cjitter	when a search does no better than sampling

The 1997 thesis derives backpropagation as gradient descent on an error surface; the 2001 placer, deployed in industry, is the same descent without a gradient, one label moved a random step and the step kept when the total cost fell. That operator, a proposal that moves one object and is judged on the summed energy, is the **block** field of the library, and the directional test of equation (2) is the same proposal made exhaustive: every direction tried, none accepted, the answer read off. The present paper starts from where a person placed the objects and reads which terms pull.

4 The corpora

Three collections of diagrams whose node coordinates a person chose, from three communities with different authoring tools, all redistributable.

corpus	licence	files	parsed	15–40 nodes	used	median m/n
WikiPathways GPML, <i>Homo sapiens</i>	CC0	1,016	571	305	305	1.06
Reactome SBGN-ML	CC0	1,382	710	248	248	1.04
BPMN Academic Initiative	academic	29,810	21,784	6,723	300	1.11

The layouts are working documents: Reactome diagrams are laid out by curators [27], WikiPathways pathways are drawn in PathVisio [28], and the BPMN models are students’ work in Signavio [29, 30]. Nobody was asked to draw nicely. The three tools share no layout code. Two WikiPathways files (WP5037, WP3391) carry coordinates converted from Reactome [31] and are dropped.

The BPMN archive is mixed: of the 21,784 models the parser reads, 13,435 have every edge a BPMN flow or association and the rest are EPC, Petri net and UML models, whose flow elements carry stencils outside the connector set and so become degree-2 nodes under the node rules. Only the 13,435 enter, 6,723 of them in the band, and the first 300 by model id are used; the two biological corpora are used whole. An earlier version of this study took 147 models without the filter, two fifths of them read as bipartite graphs.

A diagram enters as its largest connected component (194 of the 248 Reactome graphs and 120 of the 300 BPMN graphs in the band are the component of a larger diagram; no WikiPathways pathway in the band is); container glyphs (SBGN compartments, BPMN pools and lanes) are not nodes; SBGN process glyphs are nodes, so the Reactome graphs are bipartite and their m/n is the process graph’s; each layout is rescaled so that its bounding box fits the unit square, box sizes scaled with it. The directional test runs on 15 to 40 nodes; the climber pilot used 10 to 400. Diagrams are not deduplicated: the WikiPathways set is *Homo sapiens* only, so it holds no ortholog copies, and no check was made for BPMN models submitted twice. The 41 to 100 node band was not run.

How the coordinates were made matters for the alignment criterion. Of the raw coordinates, 17% are integers in WikiPathways, 67% in Reactome and 63% in BPMN (44% multiples of five); the fraction of boxes sharing an exact x with another box is 0.37, 0.06 and 0.42, an exact y 0.29, 0.12 and 0.76. In WikiPathways exact alignment is twice as common as integer coordinates, so it is the person’s act and not a rounding grid’s; BPMN authors work on a grid inside lanes, which is where the y figure of 0.76 comes from; Reactome’s curation tool rounds and its authors align little.

5 The method

Terms. With n nodes, m edges, box i of width w_i and height h_i , L the reference length of Section 2, fitted to the layout under test:

term	definition	zero when
T_C crossings	proper crossings between segments of two routes with no shared endpoint, $/m$	planar
T_O overlap	overlap area of boxes / total box area	no overlap
T_L length uniformity	mean over edges of $((\ell - L)/L)^2$	all edges of length L
T_R orthogonality	mean over edges of the length-weighted mean over its segments of $\min(\Delta x , \Delta y)/(\Delta x + \Delta y)$	routes axis-aligned
T_A alignment	1 – fraction of nodes in an alignment of ≥ 3 (tol. 0.005)	every node aligned
T_N node–edge	squared shortfall of node to foreign route distance below $(w + h)/4$	none within
T_F flow	mean over directed edges of the backward component along the reading direction u , over length	no edge runs back

The base energy is $T_C + T_O + T_L$ with weights $(1, 1, 1)$, the form of the simulated-annealing layout of Davidson and Harel [2] and of the multicriteria tradition since [24]. The field’s other

default is stress [25, 26], $\sum_{i < j} (\|x_i - x_j\| - L \delta_{ij})^2 / \delta_{ij}^2$ with δ_{ij} the graph distance, which force-directed tools and **neato** minimise and which Dwyer et al. [12] found human choices track; it is the second base energy. T_R , T_A and T_N are the candidates for what the base energy misses. The alignment definition above is HOLA’s gridiness [8], the published form, a step function of position. Three continuous forms are the sensitivities: A1, per node the distance to the nearest other node along whichever axis is nearer, which has a corner at exact alignment; A2, rows and columns priced separately; A3, a Gaussian kernel of width $s = 0.02$, smooth. The corner is what holds a box: A1 holds 0.52 of WikiPathways boxes and A3 holds 0.20.

Routes. An edge is the polyline the file stores: the attachment point on the source box, the interior waypoints, the attachment point on the target box; where the file stores nothing it is the chord between the centres. Crossings, orthogonality and node–edge separation are evaluated on the route; length and stress on the centres. When a node moves its attachment points move with it and the waypoints stay. The attachment point matters: of the 2,688 BPMN routes that carry an interior waypoint, 0.27 are axis-aligned at every segment when the route is closed at the box centres and 0.88 when it is closed at the stored attachments. What each format stores, as a fraction of the edges in the band:

corpus	edges	with a waypoint	by what the file stores
WikiPathways	8,253	0.291	chord 0.587, anchor 0.225, elbow 0.154, curved 0.026, segmented 0.008
Reactome	6,827	0.069	chord 0.931, stored 0.069
BPMN	7,910	0.327	chord 0.673, stored 0.327

GPML’s Elbow and Curved connectors store their endpoints only and PathVisio computes the bends at draw time, so those routes read as chords; a quarter of WikiPathways edges end on an anchor of another interaction, where the person drew no segment to the node and the last segment is manufactured. Only the BPMN routes are orthogonal: of the WikiPathways and Reactome routes with a waypoint, 0.009 and 0.007 are axis-aligned at every segment, the biological formats drawing diagonal segments. The chords are the sensitivity.

Reading direction. T_F needs a direction u . It is one of the four axis directions, the one under which T_F is least for the layout as loaded, fixed before any move, the same rule as for L . Undirected edge kinds (GPML graphical lines, BPMN undirected associations) do not enter it.

Directional test. Radius $d \in \{0.005, 0.01, 0.02, 0.05\}$ of the drawing width, sixteen directions, every node, each term alone and the sums of the tables. Primary cell $d = 0.02$. The directions are exact to double precision (a series evaluated in the first quadrant and rotated), and a move that lowers the energy by less than 10^{-12} of its value is a tie; an earlier build with directions off by up to 4×10^{-6} counted the shift that error made as an improvement and read alignment lower by up to 0.04. Deterministic: no seed, no budget. Every median carries a 95% bootstrap interval over diagrams (1000 resamples).

Tool controls. The same graphs with the same box sizes and edge directions laid out by **neato** (`-Gstart=1`, so the file reproduces), by **neato** with `-Goverlap=false` (prism), by **dot**, and by ELK layered 0.12.0 (**elkjs**, direction RIGHT, every other option at its default), the engine the BPMN and KIELER editors run; a check in the binary confirms each control is the same graphs.

Descent, secondary. Through the library: hill climbing from the hand layout, one node per proposal, first step $d/2$ per coordinate, the step halved after 200 rejected proposals, each node kept within a disc of radius d about its start by the repair, 4000 energy evaluations, 15 seeds per instance; and uniform sampling in the same cap at the same budget on the same seeds, the matched-budget control. Reported per instance: the fraction of the cap used (mean over nodes

of the distance moved over d), the fraction of energy removed, for the climber and for the control, and the number of seeds on which the climber ends lower than the control with its exact sign test. The climber pilot used a step of 0.1, five times the cap, so that nearly every proposal landed on the cap circle; the base-energy result did not depend on it (98.5 to 95.9% of the cap used as the step fell from 0.1 to 0.005) and the alignment result did (47 to 18%), which is why the directional test is primary. The converged reference is the same climber uncapped, 60,000 evaluations, with q_d measured where it stops.

Fitting. Weights on the simplex, the linear program of Section 2 over every node and direction of a corpus (about 120,000 inequalities, HiGHS), the residual its value per node; because the residual depends on the terms’ scales and q does not, q under the fitted weights is reported beside it, on the fitting diagrams and held out under 5×2 cross-validation (five seeded halvings, each half fitted and scored on the other). Inverse optimisation is degenerate whenever a term is satisfied exactly: overlap holds every node, so any weight vector with all its weight on overlap has residual zero and the program returns it. The informative output is the sweep, q as the weight on one term grows from 0 with the rest on crossings and overlap, which is the test of whether any positive weight on that term can be carried.

Statistics. The unit of inference is the diagram. Per corpus, the paired comparison of q_d between the hand layout and a tool control is the Wilcoxon signed-rank test, zeros dropped, ties by midranks, exact on the observed ranks up to 50 non-zero pairs and the normal approximation with continuity and tie corrections above, with the exact sign test beside it and the Hodges–Lehmann shift with its distribution-free 95% interval; the family is the terms compared against one control within one corpus, Holm-corrected. The script carries 630 self-checks against enumeration and against R. Equivalence claims (“no reweighting repairs it”) are read off the sweep, never off a failure to reject.

6 Measurements

Every number in this section is from one battery on the corpora of Section 4, run 2026-08-23 on commit 1987633 (`example/diagrams/results/run_all.sh`; versions, seeds and corpus hashes in `results/manifest.txt`). The directional test is one deterministic pass over the whole band on the stored routes with the fitted L .

Each criterion alone. $q_{0.02}$ on the hand layout, with its 95% bootstrap interval over diagrams, and on the three tool layouts of the same graphs; in parentheses the median value of the term at the layout, 0 when satisfied.

corpus	energy	hand	neato	prism	dot	elk
WikiPathways	crossings alone	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.029)	1.00 (0.036)
	overlap alone	1.00 [1.00, 1.00] (0.000)	0.60 (0.030)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00 [0.00, 0.00] (0.219)	0.24 (0.011)	0.19 (0.014)	0.00 (0.241)	0.00 (0.127)
	stress alone	0.00 [0.00, 0.00] (0.217)	1.00 (0.042)	0.94 (0.044)	0.00 (0.201)	0.00 (0.241)
	orthogonality alone	0.19 [0.18, 0.21] (0.157)	0.00 (0.280)	0.00 (0.281)	0.19 (0.244)	0.19 (0.158)
	alignment A1 alone	0.52 [0.48, 0.54] (0.003)	0.06 (0.011)	0.07 (0.010)	1.00 (0.000)	0.57 (0.001)
	alignment A3 alone	0.20 [0.18, 0.23] (0.284)	0.03 (0.446)	0.03 (0.451)	0.45 (0.204)	0.35 (0.176)
	gridiness alone	0.87 [0.84, 0.89] (0.424)	0.78 (1.000)	0.79 (1.000)	1.00 (0.125)	0.76 (0.200)
	node-edge alone	0.88 [0.86, 0.90] (0.004)	0.89 (0.000)	1.00 (0.000)	0.78 (0.011)	0.74 (0.016)
Reactome	flow alone	0.59 [0.56, 0.62] (0.131)	0.50 (0.211)	0.50 (0.211)	1.00 (0.000)	1.00 (0.000)
	crossings alone	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.056)	1.00 (0.066)
	overlap alone	1.00 [1.00, 1.00] (0.000)	0.76 (0.022)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00 [0.00, 0.00] (0.234)	0.35 (0.005)	0.26 (0.009)	0.00 (0.232)	0.00 (0.205)
	stress alone	0.00 [0.00, 0.00] (0.212)	1.00 (0.034)	0.85 (0.042)	0.00 (0.177)	0.00 (0.197)
	orthogonality alone	0.17 [0.15, 0.18] (0.193)	0.00 (0.277)	0.00 (0.277)	0.15 (0.256)	0.12 (0.203)
	alignment A1 alone	0.21 [0.19, 0.24] (0.004)	0.06 (0.009)	0.06 (0.010)	1.00 (0.000)	0.42 (0.002)
	alignment A3 alone	0.10 [0.08, 0.12] (0.300)	0.03 (0.430)	0.03 (0.428)	0.39 (0.206)	0.31 (0.213)
	gridiness alone	0.65 [0.62, 0.68] (0.640)	0.77 (1.000)	0.75 (1.000)	0.95 (0.176)	0.74 (0.306)
BPMN	node-edge alone	1.00 [0.94, 1.00] (0.000)	0.93 (0.000)	1.00 (0.000)	0.80 (0.007)	0.75 (0.021)
	flow alone	0.62 [0.60, 0.67] (0.096)	0.45 (0.217)	0.44 (0.222)	1.00 (0.000)	1.00 (0.000)
	crossings alone	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	overlap alone	1.00 [1.00, 1.00] (0.000)	0.41 (0.073)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00 [0.00, 0.00] (0.291)	0.42 (0.006)	0.27 (0.014)	0.00 (0.250)	0.00 (0.273)
	stress alone	0.00 [0.00, 0.00] (0.203)	1.00 (0.017)	0.90 (0.024)	0.00 (0.169)	0.00 (0.190)
	orthogonality alone	0.73 [0.70, 0.75] (0.011)	0.00 (0.280)	0.00 (0.278)	0.26 (0.167)	0.25 (0.139)
	alignment A1 alone	0.91 [0.87, 0.97] (0.000)	0.06 (0.011)	0.06 (0.011)	0.89 (0.000)	0.62 (0.001)
	alignment A3 alone	0.31 [0.29, 0.33] (0.207)	0.00 (0.453)	0.00 (0.452)	0.19 (0.168)	0.20 (0.155)
	gridiness alone	0.95 [0.94, 0.97] (0.176)	0.78 (1.000)	0.78 (1.000)	0.79 (0.219)	0.69 (0.250)
	node-edge alone	1.00 [1.00, 1.00] (0.000)	1.00 (0.000)	1.00 (0.000)	0.82 (0.003)	0.76 (0.016)
	flow alone	0.85 [0.82, 0.87] (0.045)	0.62 (0.121)	0.60 (0.123)	1.00 (0.000)	1.00 (0.000)

The sums. The same test under the summed energies, equal weights.

corpus	energy	hand	neato	prism	dot	elk
WikiPathways	C+O	1.00 [1.00, 1.00]	0.60	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.17	0.19	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.60	0.94	0.00	0.00
	C+O+R	0.21 [0.19, 0.23]	0.00	0.00	0.19	0.22
	C+O+A1	0.51 [0.47, 0.53]	0.04	0.07	1.00	0.57
	C+O+A3	0.24 [0.22, 0.27]	0.03	0.03	0.50	0.41
	C+O+grid	0.84 [0.81, 0.87]	0.45	0.79	0.96	0.74
	C+O+N	0.86 [0.83, 0.88]	0.55	1.00	0.78	0.74
	C+O+F	0.56 [0.53, 0.59]	0.30	0.50	1.00	1.00
Reactome	C+O	1.00 [1.00, 1.00]	0.76	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.33	0.27	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.76	0.89	0.00	0.00
	C+O+R	0.17 [0.16, 0.19]	0.00	0.00	0.14	0.13
	C+O+A1	0.22 [0.19, 0.25]	0.05	0.07	1.00	0.43
	C+O+A3	0.11 [0.09, 0.12]	0.03	0.03	0.42	0.33
	C+O+grid	0.64 [0.61, 0.67]	0.52	0.75	0.94	0.73
	C+O+N	0.94 [0.91, 0.95]	0.71	1.00	0.80	0.75
	C+O+F	0.61 [0.58, 0.65]	0.33	0.44	1.00	1.00
BPMN	C+O	1.00 [1.00, 1.00]	0.41	1.00	1.00	1.00
	C+O+L	0.00 [0.00, 0.00]	0.24	0.29	0.00	0.00
	C+O+S	0.00 [0.00, 0.00]	0.43	0.92	0.00	0.00
	C+O+R	0.73 [0.71, 0.75]	0.00	0.00	0.28	0.40
	C+O+A1	0.89 [0.86, 0.95]	0.04	0.09	0.88	0.68
	C+O+A3	0.33 [0.30, 0.37]	0.03	0.03	0.33	0.45
	C+O+grid	0.95 [0.93, 0.96]	0.28	0.78	0.75	0.67
	C+O+N	1.00 [1.00, 1.00]	0.40	1.00	0.82	0.76
	C+O+F	0.84 [0.81, 0.87]	0.26	0.61	1.00	1.00

Interpretation of the sums. Read down a hand column. Overlap holds every box at value zero; so it does in every tool layout that removes overlaps, and **neato** without removal is the only layout it does not hold, so crossings and overlap together separate a hand layout from that

configuration and from nothing else. Length and stress hold no hand-placed box, and **neato**'s stress layouts are held under stress at 1.00, so the zero is the layout's and not the test's; **dot**, which is layered and prices no distance, scores 0.00 on both as the hand layouts do. At the hand layout the three base terms share the energy as length 0.99, crossings 0.01, overlap 0.00, and $C + O + L$ inherits the length term's zero. Alignment A1 holds 0.52, 0.21 and 0.91 of hand-placed boxes, 0.06 of **neato**'s and, because a layer is a row, 1.00, 1.00 and 0.89 of **dot**'s and 0.57, 0.42 and 0.62 of ELK's; the gridiness value puts 58%, 36% and 82% of hand-placed boxes in an alignment of three or more, none of **neato**'s, 88%, 82% and 78% of **dot**'s. The gridiness q of 0.77 to 0.78 on **neato**'s layouts, at value 1.000, is the step term's flatness, which is why alignment is compared on the value. Node-edge separation is nearly satisfied in every layout (0.000 to 0.021) and holds 0.74 to 1.00 for the same reason. Crossings hold every box at this radius; the median value on the routes is 0.000 by hand in every corpus and 0.000 for **neato**, 0.029, 0.056 and 0.000 for **dot**, 0.036, 0.066 and 0.000 for ELK. Under flow **dot** and ELK are held at 1.00 with the term at 0.000 in every corpus; the hand layouts hold 0.59, 0.62 and 0.85 at 0.131, 0.096 and 0.045, and **neato**, which knows no direction, 0.44 to 0.62 at 0.121 to 0.222.

Radius. q_d at $d \in \{0.005, 0.01, 0.02, 0.05\}$, hand layouts, then **neato**'s (`profile.py -sweep`):

energy	WikiPathways				Reactome				BPMN			
	.005	.01	.02	.05	.005	.01	.02	.05	.005	.01	.02	.05
overlap, hand	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
length, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
stress, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
alignment A1, hand	0.52	0.54	0.52	0.44	0.12	0.17	0.21	0.19	0.90	0.91	0.91	0.87
gridiness, hand	0.94	0.91	0.87	0.75	0.82	0.75	0.65	0.57	1.00	1.00	0.95	0.88
flow, hand	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.63	0.85	0.85	0.85	0.85
overlap, neato	0.60	0.60	0.60	0.60	0.75	0.76	0.76	0.76	0.41	0.41	0.41	0.43
length, neato	0.04	0.12	0.24	0.47	0.07	0.19	0.35	0.58	0.08	0.22	0.42	0.72
stress, neato	0.90	0.96	1.00	1.00	0.96	1.00	1.00	1.00	0.94	0.97	1.00	1.00
alignment A1, neato	0.04	0.05	0.06	0.06	0.03	0.05	0.06	0.07	0.03	0.05	0.06	0.07
gridiness, neato	0.93	0.86	0.78	0.65	0.90	0.83	0.77	0.61	0.92	0.85	0.78	0.67
flow, neato	0.50	0.50	0.50	0.50	0.45	0.45	0.45	0.45	0.62	0.62	0.62	0.62

The hand verdicts do not move with the radius. **neato**'s under length do: a smooth term near a minimum with gradient g and curvature λ holds a node when $|g| < d\lambda/2$, so a layout that is nearly a minimum is held at a large radius and free at a small one, and **neato**'s layouts, which minimise stress and not length uniformity, read as nearly uniform at $d = 0.05$ (0.47 to 0.70) and not at $d = 0.005$ (0.04 to 0.08). A hand layout at 0.00 across the range is far from the minimum at every scale the test resolves. Gridiness falls with d in every layout because a larger move reaches an alignment from further away; flow, a projection, does not move with d at all. ELK's block (`results/sweep.md`) reads as **dot**'s: length and stress 0.00 at every radius, flow 1.00 at every radius.

Reference length. Under the median edge length instead of the fitted L , **neato**'s stress layouts score 0.07, 0.08 and 0.12 under stress and the hand layouts 0.00; under length the hand layouts score 0.03, 0.04 and 0.05 and **neato**'s 0.26, 0.41 and 0.45. Under $L = 1/\sqrt{N}$ no box of any layout is held under either term, the scale being wrong for every layout. The hand verdicts do not depend on the choice; the control's does, which is why the fit is primary (Section 2).

Descent and chance. The secondary estimand, per corpus and energy: the median fraction of the cap the capped climber uses and of the energy it removes, the same for uniform sampling in the cap at the same budget and seeds, the share of diagrams where the climber ends lower on at least 12 of 15 seeds, and $q_{0.02}$ where the uncapped climber stops, the converged reference.

corpus	energy	cap used		energy removed		$\geq 12/15$	q conv.
		climb	random	climb	random		
WikiPathways	C+O+L	0.99	0.76	0.28	0.16	0.88	1.00
	C+O+A1	0.39	0.74	0.71	0.17	0.75	1.00
	stress	1.00	0.77	0.12	0.05	1.00	1.00
	C+O+F	0.44	0.76	0.27	0.18	0.77	1.00
Reactome	C+O+L	1.00	0.76	0.31	0.16	0.93	1.00
	C+O+A1	0.56	0.74	0.99	0.37	0.80	1.00
	stress	1.00	0.77	0.13	0.06	1.00	1.00
	C+O+F	0.36	0.75	0.32	0.22	0.71	1.00
BPMN	C+O+L	1.00	0.77	0.26	0.12	0.99	1.00
	C+O+A1	0.11	0.73	0.24	0.00	0.95	1.00
	stress	1.00	0.77	0.14	0.07	1.00	1.00
	C+O+F	0.16	0.74	0.09	0.04	0.66	1.00

Under the distance energies the climber walks every node to the cap circle (0.99 to 1.00 of d) and is still removing energy when the budget ends; under C+O+A1 it moves 0.11 to 0.56 of the cap, the misalignments being nearer than d . The climber beats the control on all 15 seeds, which survives Holm over the corpus (exact sign $p = 3.1 \times 10^{-5}$ against 0.05/305), on 0.81 to 0.99 of diagrams under C+O+L and on all of them under stress. The converged reference’s median q is 1.00 for every energy and corpus, the test’s power on the sums; the fraction of diagrams whose reference stops below 0.9 is at most 0.02 everywhere but C+O+F (0.05, 0.07 and 0.13), the flow term’s plateaus.

Reweighting. The linear program is degenerate as Section 2 says it must be: on the two biological corpora it puts all weight on overlap (residual per node 0.0001 and 0.0000; held-out q 0.987 [0.982, 0.993] and 0.994 [0.990, 0.998]), on BPMN all weight on crossings (0.0009; 0.992 [0.979, 0.998]), and the same answer comes back whether it is offered three terms or all eight. Confined to length and stress it puts everything on stress and holds 0.003 to 0.011 of nodes: no distance-only weighting holds a hand layout. Confined to the four terms no layout satisfies exactly (orthogonality, alignment, node–edge, flow) it splits weight between flow and orthogonality and holds 0.22 to 0.55, except Reactome, where it rides the node–edge term and its held-out q swings from 0.10 to 0.90 across splits. The informative output is the sweep, the fraction of the corpus’s nodes held as a weight w on one term joins crossings and overlap at equal weight:

term added to C+O	WikiPathways		Reactome		BPMN	
	$w=0.001$	$w=1$	$w=0.001$	$w=1$	$w=0.001$	$w=1$
none ($w = 0$)	0.94		0.97		0.98	
length	0.008	0.004	0.010	0.007	0.012	0.009
stress	0.005	0.003	0.008	0.006	0.014	0.011
alignment A1	0.286	0.229	0.129	0.114	0.382	0.334
flow	0.576	0.606	0.613	0.625	0.797	0.810

The fall is a step, not a slope: whatever weight length or stress carries, 0.001 or 1, the sum holds under 0.015 of the nodes, because at almost every node some direction lowers the distance term and nothing else moves. Alignment and flow can carry weight: the criteria people partly satisfy survive in the sum at the fraction they are satisfied.

Routes. The tables above are on the stored routes; the chords are the sensitivity, `chords.md` beside the tables in the repository. Closing every edge at the box centres moves no verdict: crossings still hold every box (the WikiPathways value rises from 0.000 to 0.038, one crossing per 26 edges, the anchors’ work), orthogonality falls from 0.19, 0.17 and 0.73 to 0.16, 0.05 and 0.40, and node–edge separation from 0.88, 1.00 and 1.00 to 0.76, 0.85 and 0.87. The drawn geometry only raises the hand layout’s standing on the criteria it touches.

Lanes. 82% of BPMN boxes sit in a lane or pool, and a horizontal lane constrains y , so the A1 hold of 0.91 could be the container’s (`results/lanes.py`). Per held box, the alignment that holds it: 30% a column, which no horizontal lane imposes; 13% a row shared with a box of another lane or of no lane, which no lane imposes either; 57% a row within one lane. Those lanes are a median 6.3 box heights tall (quartiles 3.7 and 9.4; 3% under two box heights), so the row placement inside them was chosen, not forced. At most the choice of row count is the lane’s; where the box sits on that row, and the 43% held across or against the lanes, is the person’s.

The paired tests. Per corpus, hand against `dot` on each criterion alone: the Hodges–Lehmann shift of $q_{0.02}$ with its 95% interval, zeros dropped, and the Holm-adjusted signed-rank p within the corpus’s family (`analyse.py`; the same families against `neato`, `prism` and `ELK` are in `results/tests/`).

energy	WikiPathways		Reactome		BPMN	
	HL [95%]	p	HL [95%]	p	HL [95%]	p
crossings	−0.04 [−0.07, −0.01]	0.016	+0.02 [−0.01, +0.05]	0.30	+0.10 [+0.08, +0.12]	<1e-4
overlap	−0.14 [−0.18, −0.11]	<1e-4	−0.07 [−0.10, −0.06]	0.0024	−0.22 [−0.29, −0.11]	<1e-4
length	−0.01 [−0.03, +0.00]	0.27	−0.03 [−0.05, −0.01]	0.017	−0.09 [−0.12, −0.06]	<1e-4
stress	−0.03 [−0.04, −0.00]	0.022	−0.03 [−0.05, −0.01]	0.0048	−0.08 [−0.11, −0.05]	<1e-4
orthogonality	+0.01 [−0.00, +0.04]	0.27	+0.02 [+0.00, +0.04]	0.036	+0.43 [+0.41, +0.45]	<1e-4
alignment A1	−0.46 [−0.49, −0.43]	<1e-4	−0.70 [−0.73, −0.67]	<1e-4	+0.04 [−0.01, +0.09]	0.14
alignment A3	−0.25 [−0.28, −0.22]	<1e-4	−0.28 [−0.31, −0.26]	<1e-4	+0.13 [+0.10, +0.16]	<1e-4
gridiness	−0.13 [−0.15, −0.11]	<1e-4	−0.29 [−0.32, −0.26]	<1e-4	+0.21 [+0.17, +0.25]	<1e-4
node–edge	+0.13 [+0.10, +0.16]	<1e-4	+0.18 [+0.15, +0.21]	<1e-4	+0.26 [+0.23, +0.28]	<1e-4
flow	−0.40 [−0.42, −0.38]	<1e-4	−0.33 [−0.35, −0.31]	<1e-4	−0.17 [−0.19, −0.14]	<1e-4

The shifts under crossings, overlap, length and stress are distinguishable but confined to the minority of diagrams where the pair differs at all. The large shifts are the profile’s gaps: the hand layout sits below `dot` on alignment in the biological corpora (−0.46 and −0.70) and level with it on BPMN ($p = 0.14$), below it on flow everywhere (−0.17 to −0.40), and above it on node–edge separation and, in BPMN, on orthogonality (+0.43), the criteria `dot` trades away.

7 What else could produce this

1. **The editor snapped to a grid**, so alignment is the tool’s. WikiPathways has 17% integer coordinates and 0.37 of boxes exactly aligned in x , so there it is the person’s. BPMN is snapped (63% integers, 44% multiples of five). The half-grid jitter test, perturb every box by half a cell and repeat, was not run.
2. **The reference length.** A distance term’s verdict could be its scale’s. Tested: under the median edge length and under $1/\sqrt{N}$ the hand layouts are held under length and stress on 0.00 to 0.06 of boxes, as under the fit; only the control moves.
3. **Straight segments where the person drew polylines.** Excluded: the primary tables are on the stored routes and the Routes paragraph gives the chords beside them; no verdict moves.
4. **The person minimised a constrained energy whose constraints are absent:** reading direction, layers, compartments, lanes. Tested both ways. The flow term is in the tables: it holds 0.59 to 0.85 of hand-placed boxes, so reading direction is a criterion people partly satisfy, not the missing constraint that would explain the rest. The Lanes paragraph bounds the containers’ share of the BPMN alignment: 43% of the held boxes are held by an alignment no horizontal lane can impose, and the lanes of the rest leave a median 6.3 box heights of room. Containment as a hard constraint was not added.

5. **The radius.** At $d = 0.02$ on a thousand-pixel canvas the move is twenty pixels, about one grid step; a person cannot be stationary below the resolution they control. The Radius table answers it: the hand verdicts are the same at every d from 0.005 to 0.05. The table at d in units of box width and of median edge length was not run.
6. **Selection.** Near-trees make overlap and crossings easy. Tested: over diagrams (`sparsity.py`), q under crossings does track density (Spearman ρ against m/n : -0.37 , -0.34 and -0.11 by hand, -0.48 , -0.30 and -0.41 for `dot`), so “crossings hold every box” is partly the trees’ gift, given to every layout of the same graphs alike. q under overlap does not (ρ $+0.02$, -0.06 and $+0.12$), and length and stress are 0.00 in the sparse and the dense half both. Sparsity explains none of the hand-tool differences, which are differences on the same graphs.
7. **The tools are the wrong tools.** `dot` and `neato` are not what any of the three communities’ editors run. Excluded: ELK layered 0.12.0, the engine the BPMN and KIELER editors run, is the fourth control in every table, and its profile is `dot`’s: length and stress 0.00, flow 1.00 at value 0.000, alignment 0.42 to 0.62 where `dot` reaches 0.89 to 1.00.

8 Related work

Aesthetic criteria and their weighting go back to Tamassia, Di Battista and Batini [3] and were made an explicit energy by Davidson and Harel [2]; Purchase [4] gave the criteria their metrics, and that list has no alignment metric. Whether people agree with the criteria has been asked three ways. By having people draw: van Ham and Rogowitz [11] found hand layouts with fewer crossings than the force-directed algorithm produced; Dwyer et al. [12] had 32 participants lay out one graph and 194 judges rank the results against three automatic layouts, and found the judges’ choices track stress; Purchase, Pilcher and Plimmer [20] had participants draw from adjacency lists and found crossing removal the strongest aesthetic and alignment to a grid important; Kieffer et al. [8] collected 17 designers’ layouts of 8 small graphs, ranked by 66 judges, and found their gridiness, the fraction of nodes in some alignment of three or more, correlated with rank on all 8 graphs, which is where HOLA’s alignment step comes from. Alignment as a human criterion is therefore known since 2012; what is new here is the count of how many hand-placed boxes it holds, on diagrams of 15 to 40 boxes, beside the counts for the criteria tools weight. By showing people drawings: Purchase et al. [13] ran a Turing test in which hand drawings were distinguishable from algorithmic ones and judged better; Klammler, Mchedlidze and Pak [9] fitted the weights of a linear aesthetic combination by Nelder–Mead to discriminate good from worsened layouts, without human coordinates, and wrote that human-labelled data “may become a critically important accomplishment in the graph drawing field”. By learning from examples: Masui [10] learned a weighted constraint combination from user-supplied good and bad layouts by genetic programming, and Spönemann, Duderstadt and von Hanxleden [21] adjusted weights from user selections among candidates; neither tested whether the examples were minima of the learned energy. The closest prior descent from a hand layout is Flud [14]: simulated annealing continued from crowd layouts of pathway diagrams raised an asserted five-term score, on three networks, read as a success of the hybrid; the cap, the per-term test, the reference and the controls here are what turn that observation into a measurement of the score. Van Wagoningen, Mchedlidze and Telea [15] show drawings with identical metric values that differ in quality, which is why a fitted energy with no residual would still not be the energy people use, and why the claims here stay on the negative side. Kobourov, Pupyrev and Saket [22] found crossings matter little on sparse large graphs, consistent with finding 6. Inverse optimisation, the recovery of an objective from solutions taken as optimal, is due to Ahuja and Orlin [17] for linear programs and to Keshavarz, Wang and Boyd [18] for the convex case; Chan, Lee and Terekhov [19] define the goodness-of-fit residual reported here. Matched-budget random controls are standard in

hyperparameter search [6] and were prescribed as the baseline for genetic algorithms in 1995 [7]; we have not found them in layout. No paper fitting energy weights to human-placed coordinates was found in four searches of 2024 to 2026 work; the nearest, Zhang et al. [23], learns from 64,222 pairwise preference labels, not coordinates.

9 Reproducibility

Parsers, the energy (one function per term), the directional test, the descent, the fitting scripts, the parsed corpora and every table’s script are in the repository under `example/diagrams`. The directional test is deterministic; the descents are deterministic from their seed on every platform. Every table is from the battery of 2026-08-23 on commit 1987633, run by `results/run_all.sh` followed by `results/tests.sh`, `results/lanes.py` and `results/sparsity.py`; `results/manifest.txt` records the platform, the tool versions (gcc 13.3.0, Graphviz 2.43.0, elkjs 0.12.0, HiGHS via scipy 1.18.1), the seeds and the SHA-256 of the three raw archives and of the eighteen corpus files as measured.

10 Conclusion

People draw at a minimum of overlap, exactly, as every tool that removes overlaps does. They draw nowhere near a minimum of uniform edge length or of stress, the criteria that carry the energy in force-directed tools, whose own layouts the test holds. What they do that the base energy does not price is align boxes into rows and columns, for a fifth of the boxes in Reactome, half in WikiPathways and nine tenths in BPMN, by hand where the canvas is continuous and beyond anything the lanes impose where it is not, and point directed edges the reading way, for three fifths to six sevenths of the boxes; the layered tools do both exactly. On every criterion measured here a hand layout reads as a layered layout with partial alignment and partial flow. A tool builder has two changes to make and one number to report: take the weight off edge length, add alignment, and print the fraction of boxes the energy holds when started from a layout a person accepted.

References

- [1] M. R. Garey, D. S. Johnson. Crossing Number is NP-Complete. *SIAM Journal on Algebraic and Discrete Methods* 4(3):312–316, 1983.
- [2] R. Davidson, D. Harel. Drawing Graphs Nicely Using Simulated Annealing. *ACM Transactions on Graphics* 15(4):301–331, 1996.
- [3] R. Tamassia, G. Di Battista, C. Batini. Automatic Graph Drawing and Readability of Diagrams. *IEEE Transactions on Systems, Man, and Cybernetics* 18(1):61–79, 1988.
- [4] H. C. Purchase. Metrics for Graph Drawing Aesthetics. *Journal of Visual Languages and Computing* 13(5):501–516, 2002.
- [5] S. G. Kobourov. Force-Directed Drawing Algorithms. In *Handbook of Graph Drawing and Visualization*, CRC Press, 2013.
- [6] J. Bergstra, Y. Bengio. Random Search for Hyper-Parameter Optimization. *Journal of Machine Learning Research* 13:281–305, 2012.
- [7] A. Juels, M. Wattenberg. Stochastic Hillclimbing as a Baseline Method for Evaluating Genetic Algorithms. *Advances in Neural Information Processing Systems 8* (NIPS 1995), MIT Press, 430–436, 1996.

- [8] S. Kieffer, T. Dwyer, K. Marriott, M. Wybrow. HOLA: Human-like Orthogonal Network Layout. *IEEE Transactions on Visualization and Computer Graphics* 22(1):349–358, 2016.
- [9] M. Klammler, T. Mchedlidze, A. Pak. Aesthetic Discrimination of Graph Layouts. *Journal of Graph Algorithms and Applications* 23(3):525–552, 2019.
- [10] T. Masui. Evolutionary Learning of Graph Layout Constraints from Examples. *Proc. UIST '94*, 103–108, 1994.
- [11] F. van Ham, B. Rogowitz. Perceptual Organization in User-Generated Graph Layouts. *IEEE Transactions on Visualization and Computer Graphics* 14(6):1333–1339, 2008.
- [12] T. Dwyer, B. Lee, D. Fisher, K. I. Quinn, P. Isenberg, G. Robertson, C. North. A Comparison of User-Generated and Automatic Graph Layouts. *IEEE Transactions on Visualization and Computer Graphics* 15(6):961–968, 2009.
- [13] H. C. Purchase, D. Archambault, S. Kobourov, M. Nöllenburg, S. Pupyrev, H.-Y. Wu. The Turing Test for Graph Drawing Algorithms. *Graph Drawing (GD 2020)*, LNCS 12590, 2020.
- [14] A. Bharadwaj, D. Gwizdala, Y. Kim, K. Luther, T. M. Murali. Flud: A Hybrid Crowd–Algorithm Approach for Visualizing Biological Networks. *ACM Transactions on Computer–Human Interaction* 29(1), 2022.
- [15] S. van Wagoningen, T. Mchedlidze, A. Telea. Same Quality Metrics, Different Graph Drawings. *Graph Drawing (GD 2025)*, arXiv:2508.15557, 2025.
- [16] G. Di Battista, A. Garg, G. Liotta, R. Tamassia, E. Tassinari, F. Vargiu. An Experimental Comparison of Four Graph Drawing Algorithms. *Computational Geometry* 7(5–6):303–325, 1997.
- [17] R. K. Ahuja, J. B. Orlin. Inverse Optimization. *Operations Research* 49(5):771–783, 2001.
- [18] A. Keshavarz, Y. Wang, S. Boyd. Imputing a Convex Objective Function. *IEEE International Symposium on Intelligent Control*, 613–619, 2011.
- [19] T. C. Y. Chan, T. Lee, D. Terekhov. Inverse Optimization: Closed-Form Solutions, Geometry, and Goodness of Fit. *Management Science* 65(3):1115–1135, 2019.
- [20] H. C. Purchase, C. Pilcher, B. Plimmer. Graph Drawing Aesthetics: Created by Users, Not Algorithms. *IEEE Transactions on Visualization and Computer Graphics* 18(1):81–92, 2012.
- [21] M. Spönemann, B. Duderstadt, R. von Hanxleden. Evolutionary Meta Layout of Graphs. *Diagrams 2014*, LNCS, 16–30, 2014.
- [22] S. G. Kobourov, S. Pupyrev, B. Saket. Are Crossings Important for Drawing Large Graphs? *Graph Drawing (GD 2014)*, LNCS, 234–245, 2014.
- [23] P. Zhang, X. Li, X. Wang, H.-W. Shen, Y. Hu. Beauty in the Eye of AI: Aligning LLMs and Vision Models with Human Aesthetics in Network Visualization. arXiv:2604.03417, 2026.
- [24] R. Ahmed, F. De Luca, S. Devkota, S. Kobourov, M. Li. Graph Drawing via Gradient Descent, (GD)². *Graph Drawing (GD 2020)*, LNCS, 3–17, 2020.
- [25] T. Kamada, T. Kawai. An Algorithm for Drawing General Undirected Graphs. *Information Processing Letters* 31:7–15, 1989.
- [26] E. R. Gansner, Y. Koren, Y. North. Graph Drawing by Stress Majorization. *Graph Drawing (GD 2004)*, LNCS, 239–250, 2005.

- [27] A. Fabregat et al. Reactome Diagram Viewer: Data Structures and Strategies to Boost Performance. *Bioinformatics* 34(7):1208–1214, 2018.
- [28] A. Agrawal, H. Balci, H. Hanspers et al. WikiPathways 2024: Next Generation Pathway Database. *Nucleic Acids Research* 52(D1):D679–D689, 2024.
- [29] M. Kunze, A. Luebbe, M. Weidlich, M. Weske. Towards Understanding Process Modeling: The Case of the BPM Academic Initiative. *BPMN 2011*, LNBIP, 44–58, 2011.
- [30] D. Sola, C. Warmuth, B. Schäfer, P. Badakhshan, J.-R. Rehse, T. Kampik. SAP Signavio Academic Models: A Large Process Model Dataset. *Process Mining Workshops (ICPM 2022)*, LNBIP, 453–465, 2023.
- [31] A. Bohler et al. Reactome from a WikiPathways Perspective. *PLOS Computational Biology* 12(5):e1004941, 2016.
- [32] V. Gavrilov. Backpropagation Feed-Forward Neural Networks. Seminar thesis, Open University of Israel, 1997; revised 2026, doi:10.5281/zenodo.20450525.